Proof Workout Mastery Workbook

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I have neither given nor received unauthorized assistance.

#1 (Graded) – Warm Up THM 6

Prove: For all positive integers, A, B, if A and B are odd, $A \cdot B = C$ is **odd**.

We will now prove that $A \cdot B = C$ is odd.

There exists an integer j such that $A = 2j + 1$	definition of odd
There exists an integer k such that $B = 2k + 1$	definition of odd
$C = A \cdot B$	premise
$= (2j + 1) \cdot (2k + 1)$	substitution
$= 4jk + 2j + 2k + 1$	FOIL
$= 2(2jk + j + k) + 1$	factoring
Let h be an integer such that $h = 2jk + j + k$	closure of the integers (under multiplication and addition)
$= 2(h) + 1$	substitution

- Thus we have shown that $C = 2h+1$, therefore by the definition of an odd integer, C is an odd integer.

□

#2 Prove that if x is odd, then $5x + 3$ is "…" two different ways.

- A direct proof by **applying the Warm-Up Thms** – this means just USE the RESULTS of what you have proved above.

  We will now prove that if x is an odd integer, then $5x + 3$ is **even**.

Let $5x = A$, then we know A is an odd integer.	We deduce this from Theorem 6 which states if A and B are odd integers, then $A \cdot B = C$ is also an odd integer.
Let B be an odd integer such that $B = 3$.	By definition, B is odd because 3 is an odd number.
$5x + 3 = A + B$	by substitution

• By Theorem 3, the sum of two odd integers is even. Thus, we have shown that if x is odd, then $5x + 3$ is even. $\square$

- Using the *techniques* of the warm up Thm. and the formal definitions of even and odd. (This will look a lot like your WUT proofs.)

  We will now prove that if x is an odd integer, then $5x + 3$ is **even**.

Let $x = 2m + 1$, where $m \in \mathbb{Z}_{\geq 0}$	definition of an odd number
$5x + 3 = 5(2m + 1) + 3$	by substitution
$= 10m + 5 + 3$	distribution/multiplication
$= 10m + 8$	addition
$= 2(5m + 4)$	factoring
Let n be an integer greater than zero such that $n = 5m + 4$	closure of the integers (under multiplication and addition)
$= 2(n)$	by substitution

• Thus, $5x + 3 = 2n$ when n is an integer greater than zero. Therefore by definition, $5x + 3$ is even.

$\square$

#3 Prove using a proof by contrapositive. (Be sure to use the right format.)

If $3n + 7$ is odd, then n is even (n is an integer).

We will prove this by contrapositive. Lets assume that n is odd, then we will prove $3n + 7$ is even.

Let $n = 2a + 1$ such that a is a positive integer.	By the definition of an odd number
$3n + 7 = 3(2a + 1) + 7$	by substitution
$= 6a + 3 + 7$	by distribution and multiplication
$= 6a + 10$	by addition
$= 2(3a + 5)$	factoring
Let $b = 3a + 5$, then b is an integer	closure of the integers (under multiplication and addition)
$= 2(b)$	by substitution

Thus, we have shown that if n is odd, then $3n+7$ is even. Then by contrapositive, If $3n+7$ is odd, then n is even.

$\square$

Why does this work, when a direct proof did not?

In this case, a direct proof would not allow us to begin with the clean definition of an odd or even integer. This leaves us with a poor setup to proving n is even, and we get stuck in the middle of the proof. Starting with $3n + 7 = 2k + 1$ gives you $n = \frac{2k-6}{3}$, and dividing by 3 makes it impossible to cleanly factor out a 2 to show n is even. Since the definition of a direct proof is $p \to q$ where p is the expression "$3n + 7$ is odd" and q is "n is even", a proof by contrapositive ($\neg q \to \neg p$) allows us to flip the expression, which includes the starting point of our proof to "n is odd", allowing us to start with a definition making the proof much easier.

#4 Prove for positive integers, if n^2 is even, then n is even. $\langle$careful, careful, careful$\rangle$

We will again prove this by contrapositive. Assume n is odd.

Let $n = 2a + 1$ such that a is a positive integer.	By the definition of an odd number
$n^2 = (2a+1)^2$	using substitution
$= 4a^2 + 4a + 1$	FOIL
$= 2(2a^2 + 2a) + 1$	factoring 2 out
$= 2(2a(a+1)) + 1$	factoring a out
Let $b = a + 1$ then b is an integer	closure of the integers (under addition)
Let $c = 2a(a+1) = 2ab$, then c is an integer	closure of the integers under multiplication since we have shown b is an integer
$= 2(c) + 1$	by substitution

Thus we have shown that $n^2 = 2(c) + 1$ and therefore by definition n^2 is odd. By contrapositive, since we have shown that if n is odd, then n^2 is odd, then if n^2 is even, n is also even.

$\square$

#5 Prove that $13n+3$ is even if and only if n is odd. n is an integer. (Must prove both directions, why?)

The above statement follows the same logic pattern as a proof of equivalence. Proofs of equivalence follow the tautology $(p \leftrightarrow q) \leftrightarrow (p \to q) \wedge (q \to p)$. So, in order to prove the statement above, we must prove both direction $p \to q$ and $q \to p$ where p is "$13n+3$ is even" and q is "n is odd".

Direction 1: If n is odd, then $13n+3$ is even.

We will prove this statement directly.

Let $n = 2q+1$ such that q is a positive integer.	By the definition of an odd number
$13n+3 = 13(2q+1)+3$	by substitution
$= 26q+13+3$	by distributive multiplication
$= 26q+16$	by addition
$= 2(13q+8)$	factoring 2
Let $p = 13q+8$ then p is an integer	closure of the integers (under addition)
$= 2(p)$	by substitution

Thus, we have shown that $13n+3 = 2p$ and therefore by definition, $13n+3$ is even.

Direction 2: If $13n+3$ is even, then n is odd.

We will prove this by contraposition. Assuming n is even, we have

Let $n = 2r$ such that r is a positive integer.	By the definition of an even number
$13n+3 = 13(2r)+3$	By substitution
$= 26r+3$	By multiplication
$= 26r+(1+2)$	By substitution $(1+2=3)$
$= (26r+2)+1$	By associativity
$= 2(13r+1)+1$	factoring
Let $s = 13r+1$ then s is an integer	closure of the integers (under addition)
$= 2(s)+1$	by substitution

Thus, we have shown that $13n+3 = 2s+1$ and is therefore by definition odd. So, since we have shown that if n is even then $13n+3$ is odd, then by contraposition if $13n+3$ is even, then n is odd.

Since we have shown both **Direction 1** and **Direction 2** are true, then we have proven the equivalence "$13n+3$ is even if and only if n is odd".

$\square$

#6 Assume the domain of positive integers. Prove if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Prove by contradiction (using required steps).

We will prove this by contradiction. Assume for contradiction that there exists an $n = ab$, $\neg(a \leq \sqrt{n}) \equiv a > \sqrt{n}$ and $\neg(b \leq \sqrt{n}) \equiv b > \sqrt{n}$ (by DeMorgan's law). Then if $a > \sqrt{n}$ and $b > \sqrt{n}$:

Let $a > \sqrt{n}$	As assumed in our initial statement
Let $b > \sqrt{n}$	As assumed in our initial statement
$ab > \sqrt{n} \cdot \sqrt{n}$	In this case because $a, b, n > 0$, by laws of multiplication this statement is true
$> n$	by multiplication laws

But this leads to a contradiction, because we assumed $n = ab$ but here, $ab > n$! Thus, we have shown there does not exist an $n = ab$ such that $a > \sqrt{n}$ and $b > \sqrt{n}$, and therefore if $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ is true by contradiction.

$\square$

#7 We define an exponent b of a, (a is an element of the positive real numbers, and b is an element of the natural numbers) as:

a^b is a multiplied by itself b times. Example $a^4 = a \cdot a \cdot a \cdot a$
$a^0 = 1$

We will use only the basic principles of algebra (associativity, commutativity, etc.), and the definition of exponents given above to prove the rules of exponents.

Assume a and b are elements of the positive real numbers. You must explain each step. Notice the format of this example proof and follow this structure for #8 and #9. Do not "work both sides of the equation."

Notice how this proof does not use the rule it is trying to prove. **For #7, just fill in the blanks with the correct justification of each step.**

Prove that $a^3 \cdot a^5 = a^{(3+5)}$

$a^3 \cdot a^5 = (aaa) \cdot (aaaaa)$	Def. of exponents
$= (aaaaaaaa)$	**Associative** law
$= a^8$	Definition of **exponents**
$= a^{(3+5)}$	Substitution using $8 = 3 + 5$

□

#8 Prove that $(a^3)^5 = a^{3\cdot 5}$

We will prove this directly. We have:

$$
\begin{aligned}
(a^3)^5 &= (aaa)^5 && \text{definition of exponents} \\
&= (aaa)\cdot(aaa)\cdot(aaa)\cdot(aaa)\cdot(aaa) && \text{definition of exponents (again)} \\
&= aaaaaaaaaaaaaaa && \text{associative law of multiplication} \\
&= a^{15} && \text{Definition of exponents} \\
&= a^{3\cdot 5} && \text{Substitution using } 15 = 3\cdot 5
\end{aligned}
$$

$\square$

#9 Prove that $a^5 \cdot b^5 = (ab)^5$

We will prove this statement directly. So, we have:

$$\begin{aligned}
(ab)^5 &= (aaaaa) \cdot b^5 && \text{definition of an exponent (on a)} \\
&= (aaaaa) \cdot (bbbbb) && \text{definition of an exponent (on b)} \\
&= aaaaabbbbb && \text{by multiplication} \\
&= ababababab && \text{by the law of commutativity} \\
&= (ab) \cdot (ab) \cdot (ab) \cdot (ab) \cdot (ab) && \text{by the law of associativity} \\
&= (ab)^5 && \text{definition of an exponent}
\end{aligned}$$

$\square$

#10 Considering the above, when we add the definition of exponents to the algebra, are the "rules" of exponents really a new idea or just a consequence of the existing structure of algebra? Give a thoughtful and complete answer.

Rules of exponents for what we have covered so far are just a consequence of the existing structure of algebra, or even just shorthand when you need to show a mathematical expression multiplied a given number of times. Generally, from the proofs on #7, #8, and #9, all of the steps boil down to multiplication laws, associativity, and commutativity. I do feel this only holds true if you have a^n where n is a natural number or positive integer. When looking at a^n when n becomes negative, rational, or even irrational, the idea of an exponent and proofs with exponents get much more complex and nontrivial. However, with the help of expanded theorems and the definition of logarithms, proofs of more complex exponent statements become more feasible versus only working with basic mathematical and logical axioms.